ASSIGNMENT -2 -CO2

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COURSE:	**MOBILE COMPUTING**

Q1 – Mobile Communication or Mobile Computing

1. Classification criteria

- User mobility: whether the user/device moves while using the service.
- Connection type: wired, fixed wireless, or cellular/wireless connection.
- Disconnection tolerance: whether work can continue without a network.
- Context awareness: whether location, time or surrounding conditions affect the application.
- Computing capability: whether the device stores, processes and synchronises data locally.

Wireless means that data travels without a physical cable. Mobile means that the device or user can change location while remaining useful. Therefore, wireless and mobile are not identical. Mobile communication transfers information while a user/device moves; mobile computing includes processing, storing and using information on a portable device.

2. Classification of the five cases

Case	Classification	Main problem	Reason
(a) Rugged tablet using cellular work orders	Mobile	Both; mainly mobile computing	Staff move between sites; cellular link supports communication, while the app must handle changing coverage and work-order data.
(b) Laptop connected to Ethernet at offices	Nomadic / fixed-wired at each office	Mainly mobile computing	The laptop is carried between locations, but during use it has a wired connection. It is not continuously mobile over the network.
(c) Fixed wireless router at substation	Fixed wireless	Mainly mobile communication / networking	The endpoint is fixed, but the last connection is wireless. There is no user mobility.

(d) Push-to-talk handset	Mobile	Mainly mobile communication	The handset supports real-time voice communication while the user moves.
(e) Offline meter-reading tablet	Mobile	Mainly mobile computing	It records data locally and synchronises later. Local storage and reliable synchronisation are essential.

3. Additional ambiguous case
A delivery van has a laptop connected to a Wi-Fi hotspot inside the vehicle. The van moves between towns, but the laptop may be used only when the van is parked. It is mobile as a device and uses wireless access, but the application may behave like a nomadic system if it is used only at stops. This case shows that mobility depends on both device movement and how the service is used.

4. Store-and-forward simulation
Assumption: 1,000 meter updates are generated. Each update is attempted once. Link availability is p. For immediate sending, an update is delivered only when the link is available. For store-and-forward, every update is stored and retried once per minute until the link is available. The expected additional delay is (1−p)/p minutes, assuming independent availability at each retry.

Link availability	Immediate delivered	Immediate lost	Queued delivered eventually	Queued delayed	Mean delay of queued updates
90%	900	100	1,000	100	0.11 min
60%	600	400	1,000	400	0.67 min
30%	300	700	1,000	700	2.33 min

Interpretation: immediate sending loses more updates as coverage falls. Store-and-forward avoids permanent loss but introduces delay and requires local storage, retry logic, timestamps and duplicate protection. The table is an analytical expected-value simulation; a random simulation may vary slightly.

5. Conclusion
A wireless link alone does not solve mobility problems. Mobile computing also needs offline operation, local storage, power management, synchronisation, conflict handling, security and recovery after disconnection. Cases (a) and (e) need application-level changes, especially offline queues and synchronisation. Case (d) mainly needs communication continuity and handover support.

Q2 – Sharing the Channel: FDMA vs TDMA

1. Capacity calculation
Given bandwidth per direction = 2 MHz. FDMA channel spacing = 25 kHz. TDMA carrier bandwidth = 200 kHz and each carrier has 8 time slots.

FDMA nominal channels = 2,000 kHz / 25 kHz = 80. TDMA carriers = 2,000 / 200 = 10; nominal slots = 10 × 8 = 80.

Assumptions for usable capacity: FDMA loses 10% of nominal capacity to guard-band and control allowance; TDMA reserves one of its eight slots per carrier for synchronisation/control. These are stated modelling assumptions, not universal standards values.

Scheme	Nominal capacity	Assumed overhead	Estimated usable capacity
FDMA	80 channels	10% guard/control allowance	72 channels
TDMA	80 slots	1 control slot per 8 slots on each carrier	70 traffic slots

2. Traffic comparison

Traffic type	Better expected utilisation	Reason
Continuous voice	FDMA	A continuous call occupies a stable frequency channel; there is little burst-idle time within an active call.
Bursty data (10% duty cycle)	TDMA, if slots can be scheduled/shared	A user transmits briefly and leaves other time available. Fixed dedicated slots can still waste capacity.
70% voice / 30% data	TDMA or hybrid	Voice can receive periodic slots while bursty data uses remaining slots dynamically. Traffic scheduling improves sharing.

3. Three designed traffic conditions

- Dataset A – Continuous voice: 60 users each hold a channel for a 3-minute call. Expected result: high occupancy in both systems; blocking occurs only when simultaneous calls exceed usable capacity.
- Dataset B – Bursty telemetry: 80 devices generate a short packet every 10 seconds and use about 10% duty cycle. Dedicated allocation wastes most reserved resources; pooled or dynamically assigned TDMA slots are more efficient.
- Dataset C – Edge case: 80 users generate packets at exactly the same instant (synchronised arrivals). A system with only 70 usable TDMA traffic slots must queue or block at least 10 simultaneous transmissions; FDMA with 72 free channels can serve 72 and block/queue 8.

For a dedicated-channel model, utilisation = busy resource time / total available resource time. Blocking = requests rejected because all resources are occupied. Wasted capacity = reserved but idle resource time. In the synchronised edge case, both systems experience a short burst of blocking; TDMA has 70 usable slots and FDMA has 72 assumed usable channels.

4. Design decision for campus telemetry

No. Dedicated FDMA/TDMA allocation is not justified when 90% of devices send only a 200-byte report per minute. The traffic is small and intermittent, so dedicated channels remain idle for most of each minute. A shared, scheduled packet channel with contention control or dynamic slot assignment is more suitable. Dedicated channels may be justified only where strict delay, guaranteed capacity or continuous traffic is required.

5. Conclusion
FDMA divides the frequency band; TDMA divides transmission time on carriers. Equal nominal capacity does not mean equal usable capacity because guard bands, synchronisation and control overhead differ. Traffic pattern determines actual utilisation.

Q3 – CDMA: Orthogonality, Spreading and Near-Far Problem

1. DS-CDMA operation
In direct-sequence CDMA, each user multiplies each data bit by a unique spreading-code sequence. A bit is represented by N chips. The receiver multiplies the received chips by the intended user's code and sums them. A positive sum decides bit 1; a negative sum decides bit 0.

For a spreading length N, processing gain PG = N and PG(dB) = 10 log10(N).

Code length N	Processing gain (linear)	Processing gain (dB)
8	8	9.03 dB
16	16	12.04 dB
64	64	18.06 dB

2. Predictions before execution
- Synchronous users with ideal orthogonal Walsh codes: cross-correlation is zero, so users ideally do not interfere. BER should remain very low as users increase, provided enough orthogonal codes exist and timing/power are controlled.
- Random non-orthogonal codes: cross-correlation is generally non-zero. Multiple users create interference, so BER is expected to increase as the number of users grows.
- Orthogonal codes with one user transmitting at 10 times the power: ideal synchronous orthogonality can still separate users mathematically, but practical timing errors, channel effects and imperfect power control allow the strong user to dominate weaker users.

3. Correlation and receiver
For two code vectors c_i and c_j, normalised correlation is $\rho_{ij} = (c_i \cdot c_j)/N$. Orthogonal codes have $\rho_{ij} = 0$ for $i \neq j$. A code correlated with itself has $\rho_{ii} = 1$. The receiver's despreading operation is a correlation with the wanted code.

Code length	Ideal orthogonal cross-correlation	Expected behaviour
8	0	Lower processing gain; fewer available orthogonal codes.
16	0	More spreading and more code choices than length 8.
64	0	Highest processing gain of these choices; longer chip sequence and more processing.

4. Why orthogonality can fail
Walsh-code orthogonality assumes chip-level synchronisation. If users arrive at different times, the receiver sees shifted code sequences whose cross-correlation may not be zero. Multipath propagation can also create delayed

copies. Unequal received powers create the near-far problem: a nearby strong transmitter can mask a distant weak transmitter.

Power control is essential because the base station should receive users at approximately controlled power levels. This reduces the chance that one strong user overwhelms weaker users.

5. Simultaneous-user estimate

For an ideal synchronous Walsh system, a length-N code family can provide at most N mutually orthogonal codes, with one code reserved for the receiver/reference or system use depending on the design. A practical safe estimate is fewer than N users because guard resources, pilots, imperfect synchronisation and interference reduce usable capacity. For a target BER of 10^{-3}, the actual supported number must be measured in the chosen noise, channel and power-control simulation; it cannot be fixed from code length alone.

6. Conclusion

Longer spreading codes increase processing gain and can improve interference resistance, but they do not remove the need for synchronisation and power control. Non-orthogonal codes and near-far conditions increase interference and bit errors.

Q4 – OFDMA: Giving Each User the Subcarriers They Can Use Best

1. OFDM fundamentals

For useful symbol duration T_u, subcarrier spacing $\Delta f = 1/T_u$. This spacing makes subcarriers orthogonal over the useful symbol interval. A cyclic prefix (CP) copies the end of the OFDM symbol to its beginning. It reduces inter-symbol interference caused by multipath, provided the CP is longer than the channel delay spread.

Numerical orthogonality check: for subcarriers k and m over T_u, the inner product is $(1/T_u)\int_0^{T_u} \exp(j2\pi(k-m)t/T_u)dt$. It equals 1 when k=m and 0 when k≠m. Thus three different integer-indexed subcarriers are mutually orthogonal over the useful interval.

2. Allocation approaches

Approach	Method	Expected throughput	Fairness
Fixed contiguous blocks	Each of 8 users receives 8 neighbouring subcarriers.	Moderate; some users may receive poor-fading subcarriers.	Usually moderate; equal resource count, but not equal channel quality.
Interleaved round-robin	Users receive subcarriers spread across the band in a repeating pattern.	More robust to frequency-selective fading than contiguous blocks.	Generally good because resources are distributed across frequency.
Channel-aware best-subcarrier	Assign each subcarrier to the user with the best channel gain on it.	Highest sum throughput in many fading conditions.	Can be low; users with consistently weak channels may receive few resources.

A simple rate model is $R_{uk} = \log_2(1 + P_{uk}G_{uk}/N_0)$. Total throughput is the sum of allocated rates. Jain's fairness index is $J = (\Sigma R_u)^2 / (U \times \Sigma R_u^2)$, where U is the number of users. J=1 means equal rates; smaller values mean less fairness.

3. Prediction and design decision
Channel-aware allocation should give the highest sum throughput because it assigns each subcarrier to the user who can use it most efficiently. However, it may starve users with consistently weak channels. Interleaving is a useful compromise because each user samples different parts of the frequency band.

For a lecture hall where every student must receive a minimum rate, use proportional-fair or minimum-rate-constrained allocation: first guarantee the minimum rate to every active user, then allocate remaining subcarriers to users with good channel conditions. This protects weak users while still using good channels efficiently.

4. Conclusion
OFDMA divides the available subcarriers among users. Channel-aware assignment improves total throughput, while fairness-aware constraints prevent starvation. More users increase competition for subcarriers and can reduce the gain available to each user.

Q5 – SDMA: Reusing the Same Channel in Different Directions

1. Uniform linear array
For an N-element uniform linear array with element spacing d, wavelength λ and steering angle θ_0, the array factor can be written as $AF(\theta) = \Sigma_{n=0}^{N-1} \exp[jnkd(\sin\theta - \sin\theta_0)]$, where $k=2\pi/\lambda$. The normalised magnitude is $|\sin(N\psi/2)/(N \sin(\psi/2))|$, where $\psi=kd(\sin\theta-\sin\theta_0)$.

As N increases, the main beam becomes narrower and angular separation becomes easier. Side-lobe behaviour depends on array weighting; a uniformly weighted array has noticeable side lobes, so increasing N does not automatically eliminate them.

Array elements	Expected main-lobe width	Expected spatial separation
2	Wide	Poor separation
4	Narrower	Improved
8	Still narrower	Better for separated users
16	Narrowest of these arrays	Best angular discrimination of these choices

2. SIR criterion
Define the required SIR threshold as 10 dB. When the beam is steered toward user A, the desired signal is the array response in direction A and interference is the response in direction B. For equal transmit powers, $SIR(B) = |AF(\theta A)|^2 / |AF(\theta B)|^2$. The minimum angular separation is the smallest separation for which the calculated SIR meets 10 dB. The exact angle depends on element spacing, wavelength, steering angle and array weights; therefore it must be obtained from the pattern calculation rather than assumed.

3. Practical limitations
- Users close in angle produce similar spatial signatures, so the array cannot form two clean separate beams.
- Angular spread and multipath create several arrival directions for one user.
- A moving vehicle changes its direction and channel, requiring beam tracking.
- Calibration errors, mutual coupling and side lobes reduce practical SIR.

4. Toll-plaza decision

I would not rely on SDMA as the only access method at a toll plaza because most vehicles approach from nearly the same direction. Their angular separation may be too small for the array to distinguish them. I would combine SDMA with TDMA or another scheduling method, while using SDMA when vehicles are sufficiently separated.

5. Conclusion

SDMA reuses the same time and frequency resources by separating users spatially. Larger arrays create narrower beams, but clustered users and similar directions limit the achievable separation.

Q6 – Choosing Multiple-Access Techniques for Real Deployments

1. Weighted criteria

Criterion	Weight	Reason
Spectral efficiency	25%	Important when bandwidth is limited.
Interference resistance / reliability	20%	Important for reliable communication in crowded or hostile conditions.
Support for bursty traffic	20%	Important for sensors and data traffic.
Scalability / number of users	20%	Important when many devices share the network.
Complexity and cost	15%	Important for practical deployment.

Ratings use a 1–5 scale, where 5 is best. Weighted score = Σ(weight × rating). The ratings below are engineering judgements based on the properties discussed in Q2–Q5, not universal rankings.

2. Deployment choices

Deployment	Primary technique	Useful hybrid	Reason
Wide-area cellular voice	TDMA / FDMA combination	FDMA carrier + TDMA time slots	Voice needs predictable timing and multiple users share a carrier. This combination also illustrates how real systems combine dimensions of access.
Jam-resistant tactical link	CDMA	Frequency hopping or directional SDMA where available	Spreading codes and processing gain can provide resistance to narrowband interference; power control and synchronisation remain important.
High-rate stadium downlink	OFDMA	OFDMA + SDMA/MIMO	OFDMA allocates subcarriers flexibly; spatial beams can reuse

			resources for users in different directions.
5,000 low-rate sensors	Scheduled/dynamic TDMA or packet-based shared access	TDMA + FDMA where needed	Small intermittent reports make permanently dedicated channels wasteful. Scheduling and grouping can scale better.

3. Example rating matrix

Technique	Voice	Tactical link	Stadium downlink	5,000 sensors
FDMA	4	2	2	2
TDMA	5	3	3	4
CDMA	3	5	3	3
OFDMA	4	3	5	4
SDMA	3	3	5	3

Example sensitivity check: change each criterion weight by ±20%, then renormalise weights to 100%. The broad choices remain sensible: TDMA/FDMA for predictable voice, CDMA for a spread-spectrum tactical link, OFDMA with SDMA for high-rate stadium traffic, and dynamic scheduled access for low-rate sensors. The least robust choice is CDMA for the tactical link if power control or synchronisation is poor; near-far effects can sharply reduce performance.

4. Why systems combine techniques

A single technique cannot optimise every requirement. FDMA separates users by frequency, TDMA by time, CDMA by code, OFDMA by orthogonal subcarriers and SDMA by direction. Combining them can provide frequency isolation, scheduled access, interference control and spatial reuse. The best choice depends on traffic, mobility, interference, cost and fairness.

Q7 – Diagnose, Redesign and Validate

Selected limitation: dedicated channels under bursty telemetry traffic

A. Diagnose the bottleneck (2 marks)

The dedicated-channel design reserves one channel or slot for each device even though a sensor transmits only a short report occasionally. The main counted quantity is idle reserved channel-time. As the number of sensors increases, the total reserved capacity increases even when the total transmitted data remains small. The original design wastes resources because allocation is static rather than based on actual packet arrivals.

B. Design objective (2 marks)

Objective: deliver all valid sensor reports while reducing reserved idle channel-time by at least 50% compared with permanent dedicated allocation, without allowing the report delay to exceed the application limit. For this example, set a target of delivery within 2 seconds under normal load.

C. Revised strategy (4 marks)

Use a shared request-and-grant schedule. Sensors sleep when they have no report. When a report is ready, a sensor sends a short request in a contention window. The base station groups successful requests and assigns transmission slots. A sensor releases its slot after sending the packet. A retry/backoff rule handles collisions.

- Changed property: allocation now follows actual traffic demand rather than permanently reserving a resource.
- Reduced operations: long idle reservations and unnecessary channel holding are avoided.
- New state: the base station maintains a queue of requests and the next available slot.
- Trade-off: request messages and scheduling add control overhead; simultaneous requests may collide and need retries.

Pseudocode

```
FOR each reporting interval:
    sensors with data send a short access request
    base station collects requests during the request window
    resolve collisions using random backoff and retry
    allocate available data slots to successful requests
    sensors transmit their reports in assigned slots
    release each slot after transmission
    record delay, delivered reports, collisions and occupied slots
```

D. Prediction before implementation (1 mark)

Prediction: under low and medium sensor traffic, dynamic allocation will use fewer occupied/reserved slots than permanent dedicated allocation. It may have slightly more control messages and delay because sensors first request a slot.

E. Prototype comparison (2 marks)

Illustrative comparison for 100 sensors, each producing one short report in a 60-second interval. Assume each report occupies one data slot. Permanent allocation reserves 100 slots for the full interval. Dynamic allocation uses 10 request/control slots plus one data slot per report, with reports spread over the interval.

Measure	Dedicated allocation	Dynamic request-and-grant
Resources reserved for the full interval	100 channels/slots	No permanent per-sensor reservation
Data slots used for 100 reports	100	100
Idle reserved capacity between reports	High	Low; resources are released after use
Control overhead	Low	Request and grant messages required
Collision possibility	No contention collision in dedicated model	Possible in request window; retries required

A simple utilisation illustration: if each sensor transmits for 1% of the interval, dedicated allocation has approximately 1% time utilisation per reserved channel. Dynamic allocation pools the short transmissions, so the shared resource can be reused by other sensors during the remaining time. Exact utilisation and delay should be measured by implementing both schedulers with the same random seed and arrival trace.

F. Defend the final design (2 marks)

The redesign is expected to improve resource utilisation when traffic is sparse and bursty. Its costs are request signalling, scheduling complexity and possible access collisions. Dedicated allocation may still be preferable for continuous voice, emergency traffic requiring guaranteed immediate access, or systems where request overhead is

larger than the saved capacity. The final decision should be based on measured slot occupancy, report delay, collision count and delivery ratio.

Overall conclusion

The assignment shows that multiple-access methods solve different resource-sharing problems. FDMA and TDMA divide frequency and time; CDMA separates users by spreading codes; OFDMA allocates orthogonal subcarriers; SDMA separates users by direction. Good engineering requires predicting performance, stating assumptions, measuring counted quantities, identifying bottlenecks and redesigning the system according to its traffic and service requirements.